

Practice Test 1 · Answer Key

Chemical Bonding Unit Test · 80 points

Part A · Multiple Choice (2 points each, 34 total)

Quick key: 1 B 2 D 3 A 4 C 5 B 6 D 7 C 8 A 9 D 10 C 11 A 12 B 13 D 14 C 15 A
16 B 17 C

#	Ans	Why
1	B	CaCl₂. A metal (Ca) with a non-metal (Cl). The other three are made only of non-metals, so they are covalent.
2	D	delocalized and free to move among positive ions. The "sea of electrons" model. A describes ionic bonding and B covalent bonding.
3	A	polar covalent. Two non-metals with ΔEN between 0.5 and 1.7.
4	C	32. $S 6 + O 4 \times 6 = 30$, plus 2 for the $2-$ charge = 32.
5	B	a triple bond, with one lone pair on each N. 10 valence electrons: 6 in the triple bond and one lone pair (2) on each N gives both atoms an octet.
6	D	two [K]⁺ ions and one [O]²⁻ ion with eight dots. Each K gives one electron to O. Ionic compounds are drawn as ions in brackets, not with shared pairs.
7	C	sodium chloride. Ionic: the ions can move only once the lattice melts. Copper conducts in both states; sugar and diamond in neither.
8	A	layers of ions slide past each other while the delocalized electrons keep holding them. Nothing has to break when the layers move. C explains why ionic solids are brittle.
9	D	CH₄. CH ₄ is made of separate molecules with only weak forces between them. MgO is ionic, SiO ₂ a covalent network and Fe a metal.
10	C	iron(III) oxide. Three O ²⁻ give 6-, so two Fe give 6+: Fe ³⁺ . No prefixes in ionic names.
11	A	Ca(NO₃)₂. Ca ²⁺ needs two NO ₃ ⁻ ; two of the same polyatomic ion go in parentheses. Ca ₃ N ₂ is calcium nitride.
12	B	dinitrogen tetroxide. Two non-metals: prefixes di- and tetra-, with the "a" dropped before "oxide".
13	D	a molecular covalent material such as a plastic. It does not conduct and it bends. An ionic solid would also insulate but is brittle.
14	C	trigonal pyramidal. 4 domains on N: 3 bonded atoms and 1 lone pair. Tetrahedral is the electron geometry.
15	A	180°. Two electron domains on C (each double bond counts once): linear.
16	B	CCl₄. Tetrahedral with four identical atoms, so the four C-Cl dipoles cancel.
17	C	Its bent shape means the O-H bond dipoles do not cancel. Two lone pairs on O make the molecule bent, so the two dipoles add.

Part B · Free Response (46 points)

18. Bond types and properties (8 points)

(a) 1 point for each fully correct row.

Substance	Type of bonding	Conducts as a solid?	Melting point
KBr	ionic	no	high
Cu	metallic	yes	high
candle wax	covalent (molecular)	no	low
diamond	covalent network	no	very high

(b) In the solid the **ions** are held in **fixed positions** in the lattice, so they cannot carry charge. When KBr dissolves, the ions separate and are **free to move**, so they carry the current.

1 point: ions fixed in the solid. 1 point: ions free to move in solution. No credit for "electrons move".

(c) Melting only separates the molecules from each other. The **forces between molecules** (intermolecular forces) are **weak**, so little energy is needed. The covalent bonds inside each molecule do not break.

1 point: weak forces between molecules. 1 point: covalent bonds are not broken on melting.

19. Names and formulas (10 points)

(a) 1 point each: MgBr_2 **magnesium bromide** · CuCl_2 **copper(II) chloride** · N_2O_3 **dinitrogen trioxide** · Na_2CO_3 **sodium carbonate** · PCl_5 **phosphorus pentachloride**

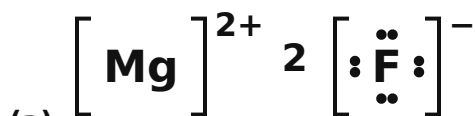
No credit for prefixes in an ionic name ("magnesium dibromide") or for a missing Roman numeral on copper.

(b) 1 point each: aluminum sulfide **Al_2S_3** · iron(III) nitrate **$\text{Fe}(\text{NO}_3)_3$** · carbon tetrachloride **CCl_4** · ammonium phosphate **$(\text{NH}_4)_3\text{PO}_4$**

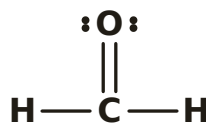
Parentheses are required around NO_3 and NH_4 . FeNO_{33} or $\text{NH}_{43}\text{PO}_4$ earns no credit.

(c) Copper can form **more than one ion** (Cu^+ and Cu^{2+}), so the name must show which charge it has. Magnesium forms only Mg^{2+} .

20. Drawing bonds (10 points)



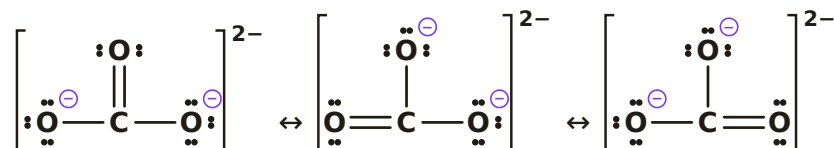
1 point: Mg has lost both valence electrons (no dots, charge 2+). 1 point: two fluoride ions, each with 8 dots. 1 point: brackets and charges on every ion.



(b) $4 + 2 \times 1 + 6 = 12$ **valence electrons.**

1 point: 12 electrons. 1 point: C in the center with a C=O double bond and two C-H bonds. 1 point: exactly 2 lone pairs on O and none on C or H.

(c) $C 4 + O 3 \times 6 + 2$ for the charge = **24 valence electrons**. The ion has **3 resonance structures**.



1 point: 24 electrons. 1 point: one C=O and two C-O bonds. 1 point: lone pairs (2 on the double-bonded O, 3 on each single-bonded O) with brackets and the 2- charge. 1 point: 3 resonance structures. Any one of the three drawings earns the structure points.

21. Molecular geometry and polarity (14 points)

Formula	Valence e ⁻	Lewis structure	Electron geometry	Molecular geometry & bond angle	3-D redraw	Bond polarity (show ΔEN)	Molecular polarity
NF ₃	26		tetrahedral (4 domains)	trigonal pyramidal ≈107°		N-F: 1.0, polar	Polar
CH ₂ O	12		trigonal planar (3 domains)	trigonal planar 120°		C=O: 1.0, polar C-H: 0.4, nonpolar	Polar

1 point per column in each row (7 columns). Redraw: NF₃ needs the lone pair on N with 1 plain line, 1 wedge and 1 dash; CH₂O is flat, so plain lines only. Bond polarity needs the ΔEN values. Accept "less than 109.5°" for NF₃.

Why NF₃ is polar: the lone pair on N makes the shape unsymmetric, so the N-F dipoles do not cancel. **Why CH₂O is polar:** the outer atoms are not all the same, so the C=O dipole is not cancelled.

22. Designing materials (4 points)

(a) **Metallic** bonding (for example stainless steel or aluminum). It **conducts heat** well because its delocalized electrons carry energy, and it is **malleable**, not brittle, so it does not crack.

1 point: metallic. 1 point: two suitable properties.

(b) **Covalent** (molecular). It is a **poor conductor of heat** (and of electricity), so the handle stays cool and safe to hold.

(c) Suitable: it does **not conduct electricity** and has a very **high melting point**. Drawback: it is **brittle**.

Both needed for the point.